Reviewer Pack — Coxeter Group Action Complexity and Resolution Proof Width I…

Entry 21b5099f35a8 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Coxeter Group Action Complexity and Resolution Proof Width Inequality

Entry ID: `21b5099f35a8` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-03 06:57:58 UTC

1. Conjecture

Field A (mathematical branch): Combinatorial Geometry (Coxeter Groups) **Field B** (complexity object): Resolution Proof Complexity

Statement:

> For every Conjunctive Normal Form (CNF) formula φ with n variables, the resolution proof width of φ is linearly correlated with the maximum number of active Coxeter reflections required to transform any initial set of vertices into a configuration representing a refutation of φ.

Rationale (proposer's reasoning):

> Coxeter groups are combinatorial structures that describe symmetries of geometric objects and have applications in geometry, algebra, and group theory. Their action on polytopes can be related to proof complexity by considering the number of symmetries needed to reach a contradiction in a logical system. This connection could expose new insights into the inherent complexity of resolution proofs.

Taxonomy category: `c003b_cumulative_entropy/SC2#c49` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `02e250349d3c8049`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The resolution proof width of a CNF formula φ with n variables is linearly correlated with the maximum number of active Coxeter reflections required for refutation if Pearson correlation coefficient (r) ≥ 0.5 and p-value ≤ 0.05 across 30 seeds.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	UNCERTAIN	1.00	SAFE	HITS
NATURAL_PROOFS	SAFE	0.70	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	SAFE
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `"Coxeter group action" AND "resolution proof complexity"` - `"resolution proof width" AND "Coxeter reflection"` - `"CNF formula" AND "active Coxeter reflections" AND "proof complexity"`

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def generate_cnf(n):
    cnf = []
    for _ in range(10 * n):  # Generate 10 clauses per variable on average
        clause = set()
        while len(clause) < n:
            lit = random.randint(-n, -1)
            if lit not in clause and -lit not in clause:
                clause.add(lit)
        cnf.append(list(clause))
    return cnf

def dpll_solve(cnf):
    def solve(lits_true, lits_false):
        if not cnf:
            return True
        unit_clause = next((c for c in cnf if len(c) == 1), None)
        if unit_clause:
            lit = unit_clause[0]
            if lit in lits_true or -lit in lits_false:
                return solve(lits_true | {lit}, lits_false)
            else:
                return False
        pure_literal = next((c for c in cnf if len(c) == 1), None)
        if not pure_literal:
            pure_literal = random.choice([l for l in range(1, n + 1)] + [-l for l in range(1, n +
        if pure_literal > 0:
            return solve(lits_true | {pure_literal}, lits_false)
        else:
            return solve(lits_true, lits_false | {-pure_literal})
    return solve(set(), set())
```

```python
def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.choice([5, 10, 15, 20, 30, 40])
    cnf = generate_cnf(n)
    resolution_width = dpll_solve(cnf)

    # Construct polytope and compute maximum number of active Coxeter reflections
    # This is a placeholder for the actual computation which is not provided in the problem statem
    max_reflections = n  # Placeholder value, replace with actual computation

    if resolution_width is None:
        return {
            "metric_name": "resolution_width",
            "metric_value": None,
            "instances_tested": 1,
            "n_max": n,
            "conjecture_holds": False,
            "counterexample": "mapping_undefined"
        }

    correlation = resolution_width / max_reflections
    return {
        "metric_name": "resolution_width",
        "metric_value": correlation,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": abs(correlation) >= 0.5,
        "counterexample": ""
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len(re
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results if r["metric_
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_f
    else:
        print(f"RESULT: INCONCLUSIVE reason=unsupported_conjecture")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4b5755eb.py", line 85, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4b5755eb.py", line 53, in run_trial
    resolution_width = dpll_solve(cnf)
                       ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4b5755eb.py", line 47, in dpll_solve
    return solve(set(), set())
           ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4b5755eb.py", line 42, in solve
    pure_literal = random.choice([l for l in range(1, n + 1)] + [-l for l in range(1, n + 1)])
                                                                                  ^
NameError: name 'n' is not defined
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data, which means that the pre-registered support condition could not be unambiguously met. | next: Re-run the test with proper error handling to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13997
2	propose	ollama_remote	glm4:latest	0	0	13758
3	preregistration	ollama_remote	glm4:latest	0	0	9454
4	novelty	ollama_remote	glm4:latest	0	0	12243
5	novelty	ollama_remote	glm4:latest	0	0	8592
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18074
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	31723
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	65189
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15155
10	judge	ollama_remote	glm4:latest	0	0	22197

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 210383 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/21b5099f35a8.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/21b5099f35a8.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/21b5099f35a8.tar.gz` (if generated)